Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Consider an ideal gas confined in an isolated closed chamber. As the gas undergoes an adiabatic expansion, the average time of collision between molecules increases as V^{q}, where V is the volume of the gas. The value of q is : $\left(\gamma=\frac{\mathrm{C}_{\mathrm{p}}}{\mathrm{C}_{\mathrm{v}}}\right)$

(A) $\frac{\gamma+1}{2}$

(B) $\frac{\gamma-1}{2}$

(C) $\frac{3\gamma+5}{6}$

(D) $\frac{3\gamma-5}{6}$

Q.2

The molar heat capacities of a mixture of two gases at constant volume is $13\mathrm{R}/6$. The ratio of number of moles of the first gas nto the second is $1:2$. The respective gases may be

(A) O_2 and N_2

(B) He and Ne

(C) He and N_2

(D) N_2 and O_2

Q.3

For a gas, difference between two specific heats is $5000\ \mathrm{J}/$ mole $^{\circ}\mathrm{C}$. If the ratio of specific heats is 1.6, the two specific heats in $\mathrm{J}/\mathrm{mole}-{}^{\circ}\mathrm{C}$ are

(A) $\mathrm{C_P}=1.33\times10^4, \mathrm{C_V}=2.66\times10^4$

(B) $C_P=13.3\times10^4, C_V=8.33\times10^3$

(C) $C_P=1.33\times10^4, C_V=8.33\times10^3$

(D) $\mathrm{C_P}=2.6\times10^4, \mathrm{C_V}=8.33\times10^4$

Q.4

The P-V diagram of a diatomic gas is a straight line passing through origin. The molar heat capacity of the gas in the process will be

(A) $4\mathrm{R}$

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(B) $2.5\mathrm{R}$

(C) $3\mathrm{R}$

(D) $\frac{4R}{3}$

Q.5

A gaseous mixture consists of $16\ \mathrm{g}$ of helium and $16\ \mathrm{g}$ of oxygen. Then what will be the ratio $\frac{C_p}{C_v}$ of the mixture ?

(A) 1.62

(B) 1.52

(C) 1.65

(D) 1.58

Q.6

The specific heats, C_p and C_V of gas of diatomic molecules, A, is given (in units of J mol $^{-1}K^{-1}$) by 29 and 22, respectively. Another gas of diatomic molecules, B, has the corresponding values 30 and $21.$ If they are treated as ideal gases, then

(A) A has a vibrational mode but B has none.

(B) A has one vibrational mode and B has two.

(C) A is rigid but B has a vibrational mode.

(D) Both A and B have a vibrational mode each.

Q.7

One mole of a gas occupies 22.4 lit at N.T.P. Calculate the difference between two molar specific heats of the gas. $\mathrm{J}=4200\ \mathrm{J/kcal}.$

(A) $1.979\mathrm{kcal/kmolK}$

(B) $2.378\mathrm{kcal/kmolK}$

(C) $4.569\mathrm{kcal/kmolK}$

(D) $3.028\mathrm{kcal/kmolK}$

Q.8

The temperature of the mixture of one mole of helium and one mole of hydrogen is increased from $0^{\circ}\mathrm{C}$ to $100^{\circ}\mathrm{C}$ at constant pressure. The amount of heat delivered will be

(A) $600\mathrm{cal}$

(B) $1200\mathrm{cal}$

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(C) 1800cal

(D) 3600cal

Q.9

The P-V diagram of a diatomic gas is a straight line passing through origin. The molar heat capacity of the gas in the process will be

(A) 4R

(B) 2.5R

(C) 3R

(D) $\frac{4R}{3}$

Q.10

The work of $146\ \text{kJ}$ is performed in order to compress one kilomole of gas adiabatically and in this process the temperature of the gas increases by 7°C. The gas is $(R = 8.3\ \text{Jmol}^{-1}\ \text{K}^{-1})$

(A) diatomic

(B) triatomic

(C) a mixture of monatomic and diatomic

(D) monatomic

Q.11

The molar heat capacities of a mixture of two gases at constant volume is $13\text{R}/6$. The ratio of number of moles of the first gas
to the second is $1:2$. The respective gases may be

(A) O_2 and N_2

(B) He and Ne

(C) He and N_2

(D) N_2 and He

Q.12

A gaseous mixture consists of $16\ \text{g}$ of helium and $16\ \text{g}$ of oxygen. Then what will be the ratio $\frac{C_p}{C_v}$ of the mixture ?

(A) 1.60

(B) 1.62

(C) 1.58

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(D) 1.64

Q.13

$4.0\ \mathrm{g}$ of a gas occupies 22.4 litres at NTP. The specific heat capacity of the gas at constant volume is $5.0 \mathrm{JK}^{-1}$. If the speed of any quantity x in this gas at NTP is $952\ \mathrm{ms}^{-1}$, then the heat capacity at constant pressure is (Take gas constant $\mathrm{R}=8.3 \mathrm{JK}^{-1}\ \mathrm{mol}^{-1}$)

(A) $7.5 \mathrm{JK}^{-1}\ \mathrm{mol}^{-1}$

(B) $7.0 \mathrm{JK}^{-1}\ \mathrm{mol}^{-1}$

(C) $8.5 \mathrm{JK}^{-1}\ \mathrm{mol}^{-1}$

(D) $8.0 \mathrm{JK}^{-1}\ \mathrm{mol}^{-1}$

Q.14

Four moles of an ideal monoatomic gas $\left(\gamma=\frac{5}{3}\right)$ are mixed with two moles of a diatomic gas $\left(\gamma=\frac{7}{5}\right)$. The value of γ of the mixture will be

(A) $\frac{7}{3}$

(B) $\frac{15}{8}$

(C) $\frac{17}{11}$

(D) $\frac{19}{12}$

Q.15

A $10\ \mathrm{kW}$ drilling machine is used to drill a bore in a small aluminium block of mass $8.0\ \mathrm{kg}$. What is the rise in temperature of the block in 2.5 minutes, assuming 50% of power is used up in heating the machine itself or lost to the surroundings?
Specific heat of aluminium $=0.91\ \mathrm{J/g^{\circ}C}$

(A) 104 $^{\circ}$C

(B) 93 $^{\circ}$C

(C) 103 $^{\circ}$C

(D) 94 $^{\circ}$C

Answers & Solutions

Q.1 Answer:

$\frac{\gamma+1}{2}$

Solution:

$$\tau=\frac{1}{\sqrt{2}\pi \mathrm{d}^2\left(\frac{\mathrm{N}}{\mathrm{V}}\right)\sqrt{\frac{3\mathrm{RT}}{\mathrm{M}}}}$$

$$\tau \propto \frac{\mathrm{V}}{\sqrt{\mathrm{T}}}$$

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As, $TV^{\gamma-1}=K$
So, $\tau \propto \mathrm{V}^{\gamma+1/2}$
Therefore, $q=\frac{\gamma+1}{2}$

Q.2 Answer:

He and N_2

Solution:

$$C_{v_{\text{mix}}}=\frac{n_1C_{v_1}+n_2C_{v_2}}{n_1+n_2}$$
$$\Rightarrow \frac{13R}{6}=\frac{n_1C_{v_1}+2n_1C_{v_2}}{n_1+2n_1} \quad \left[\because \frac{n_1}{n_2}=\frac{1}{2}\right]$$
$$\Rightarrow \frac{13R}{2}=C_{v_1}+2C_{v_2}$$

Possible values are, $C_{\mathrm{v}_1}=\frac{3\mathrm{R}}{2}, \mathrm{C}_{\mathrm{v}_2}=\frac{5\mathrm{R}}{2}$
$\therefore$ Gases are monatomic (like He) and diatomic (like N_2 **)**

Q.3 Answer:

$C_P=1.33\times10^4, C_V=8.33\times10^3$

Solution:

Given $\mathrm{C_P}-\mathrm{C_V}=5000\ \mathrm{J/mole^\circ C}$**...(i)**
$\frac{\mathrm{C_P}}{\mathrm{C_V}}=1.6$**...(ii)**
From Equation (i) \& (ii), $\Rightarrow \frac{\mathrm{C_P}}{\mathrm{C_V}}-\frac{\mathrm{C_V}}{\mathrm{C_V}}=\frac{5000}{\mathrm{C_V}}$
$\Rightarrow 1.6-1=\frac{5000}{\mathrm{C_V}}$
$\Rightarrow \mathrm{C_V}=\frac{5000}{0.6}=8.33\times10^3$
Hence $C_P=1.6C_V=1.6\times8.33\times10^3$
$\mathrm{C_P}=1.33\times10^4$

Q.4 Answer:

$3\mathrm{R}$

Solution:

P-V diagram of the gas is a straight line passing through origin. Hence $\mathrm{P}\propto\mathrm{V}$ **or** $\mathrm{PV}^{-1}=$ **constant Molar heat capacity in the process** $\mathrm{PV}^{\mathrm{x}}=$ **constant is** $C=\frac{R}{\gamma-1}+\frac{R}{1-x}$**; Here** $\gamma=1.4$ **(For diatomic gas)** $\Rightarrow C=\frac{R}{1.4-1}+\frac{R}{1+1}\Rightarrow C=3R$

Q.5 Answer:

1.62

Solution:

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For mixture of gas, $C_v = \frac{n_1 C_{v_1} + n_2 C_{v_2}}{n_1 + n_2}$

$= \frac{4 \times \frac{3}{2}R + \frac{1}{2} \times \frac{5}{2}R}{\left(4 + \frac{1}{2}\right)} = \frac{6R + \frac{5}{4}R}{\frac{9}{2}} = \frac{29R \times 2}{9 \times 4} = \frac{29R}{18}$

and $C_p = \frac{n_1 C_{p_1} + n_2 C_{p_2}}{(n_1 + n_2)} = \frac{4 \times \frac{5R}{2} + \frac{1}{2} \times \frac{7R}{2}}{\left(4 + \frac{1}{2}\right)}$

$\therefore \frac{C_p}{C_v} = \frac{47R}{18} \times \frac{18}{29R} = 1.62$

Q.6 Answer:

A has a vibrational mode but B has none.

Solution:

Here Cp and Cv of A are 29 and 22 and Cp and CV of B are 30 and 21.

$y = \mathrm{Cp/CV} = 1 + 2/\mathrm{f}$

For $A, Cp/Cv = 1 + 2/f \Rightarrow f = 6$

Molecules A has 3 translational, 2 rotational and 1 vibrational degree of freedom

For B, Cp/Cv $= 1 + 2/f \Rightarrow f = 5$

i.e., B has 3 translational and 2 rotational degrees of freedom.

Q.7 Answer:

$1.979\mathrm{kcal/kmolK}$

Solution:

$V = 22.4$ **litre** $= 22.4 \times 10^{-3}\ \mathrm{m}^3,\ \mathrm{J} = 4200\ \mathrm{J/kcal}$

by ideal gas equation for one mole of a gas, $\mathrm{R} = \frac{\mathrm{PV}}{\mathrm{T}} = \frac{1.013 \times 10^5 \times 22.4 \times 10^{-3}}{273}$

$\mathrm{C_p - C_v} = \frac{\mathrm{R}}{\mathrm{J}} = \frac{1.013 \times 10^5 \times 22.4}{273 \times 4200} = 1.979\mathrm{kcal/kmolK}$

Q.8 Answer:

$1200\mathrm{cal}$

Solution:

$(C_p)_{mix} = \frac{\mu_1 C_{p_1} + \mu_2 C_{p_2}}{\mu_1 + \mu_2}$

$\left(C_{p_1}(He) = \frac{5}{2}R \text{ and } C_{p_2}\,(H_2) = \frac{7}{2}R\right)$

$(C_p)_{\mathrm{mix}} == \frac{1 \times \frac{5}{2}R + 1 \times \frac{7}{2}R}{1+1} 3\mathrm{R} = 3 \times 2 = 6\mathrm{cal/mol.^\circ C}$

$\therefore$ **Amount of heat needed to raise the temperature from** $0^\circ\mathrm{C}$ **to** $100^\circ\mathrm{C}$

$(\Delta Q)_p = \mu C_p \Delta T = 2 \times 6 \times 100 = 1200\mathrm{cal}$

Q.9 Answer:

$3\mathrm{R}$

Solution:

P-V diagram of the gas is a straight line passing through origin. Hence $\mathrm{P} \propto \mathrm{V}$ **or** $\mathrm{PV^{-1}} =$ **constant**

Molar heat capacity in the process $\mathrm{PV^x} =$ **constant is** $C = \frac{R}{\gamma - 1} + \frac{R}{1 - x}$**; Here** $\gamma = 1.4$ **(For**

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diatomic gas) $\Rightarrow C=\frac{R}{1.4-1}+\frac{R}{1+1} \Rightarrow C=3R$

Q.10 Answer:

diatomic

Solution:

$W=\frac{nR\Delta T}{1-\gamma} \Rightarrow -146000=\frac{1000\times 8.3\times 7}{1-\gamma}$
or $1-\gamma=-\frac{58.1}{146} \Rightarrow \gamma=1+\frac{58.1}{146}=1.4$
Hence the gas is diatomic.

Q.11 Answer:

He and N_2

Solution:

$$C_{v_{\text{mix}}}=\frac{n_1C_{v_1}+n_2C_{v_2}}{n_1+n_2}$$
$$\Rightarrow \frac{13R}{6}=\frac{n_1C_{v_1}+2n_1C_{v_2}}{n_1+2n_1} \quad \left[\because \frac{n_1}{n_2}=\frac{1}{2}\right]$$
$$\Rightarrow \frac{13R}{2}=C_{v_1}+2C_{v_2}$$

Possible values are, $C_{\mathrm{v}_1}=\frac{3\mathrm{R}}{2}, \mathrm{C}_{\mathrm{v}_2}=\frac{5\mathrm{R}}{2}$
$\therefore$ Gases are monatomic (like He) and diatomic (like N_2 **)**

Q.12 Answer:

1.62

Solution:

For mixture of gas, $C_v=\frac{n_1C_{v1}+n_2C_{v2}}{n_1+n_2}$
$=\frac{4\times\frac{3}{2}R+\frac{1}{2}\times\frac{5}{2}R}{\left(4+\frac{1}{2}\right)}=\frac{6R+\frac{5}{4}R}{\frac{9}{2}}=\frac{29R\times 2}{9\times 4}=\frac{29R}{18}$
and $C_p=\frac{n_1C_{p_1}+n_2C_{p_2}}{(n_1+n_2)}=\frac{4\times\frac{5R}{2}+\frac{1}{2}\times\frac{7R}{2}}{\left(4+\frac{1}{2}\right)}$
$\therefore \frac{C_p}{C_v}=\frac{47R}{18}\times\frac{18}{29R}=1.62$

Q.13 Answer:

$8.0\mathrm{JK}^{-1}\ \mathrm{mol}^{-1}$

Solution:

Molar mass of the gas $=4\ \mathrm{g/mol}$ **Speed of any quantity** x $\mathrm{V}=\sqrt{\frac{\gamma \mathrm{RT}}{\mathrm{m}}} \Rightarrow 952=\sqrt{\frac{\gamma\times 3.3\times 273}{4\times 10^{-3}}}$
$\Rightarrow \gamma=1.6=\frac{16}{10}=\frac{8}{5}$
Also, $\gamma=\frac{\mathrm{C_P}}{\mathrm{C_V}}=\frac{8}{5}$
So, $C_P=\frac{8\times 5}{5}=8\mathrm{JK}^{-1}\ \mathrm{mol}^{-1}$

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Q.14 Answer:

$\frac{17}{11}$

Solution:

For monoatomic gas, let

C_v' **= molar specific heat at constant volume** C_p' **= molar specific heat at constant pressure For diatomic gas, let** C_v'' **= molar specific heat at constant volume**

C_p'' **= molar specific heat at constant pressure For the mixture, let** C_v **and** C_p **be the molar specific heats at constant volume and at constant pressure.**

Given $\frac{C_p'}{C_v'} = \frac{5}{3}$ **and** $\frac{C_p''}{C_v''} = \frac{7}{5}$

Also $C_p' - C_v' = R = C_p'' - C_v''$

$\therefore C_v' = \frac{3}{2}R$ **and** $C_p' = \frac{5}{2}R$

and $C_v'' = \frac{5}{2}R$ **and** $C_p'' = \frac{7}{2}R$

Suppose the temperature of the mixture is increased by dT**. The increase in the internal energy of the first gas** $= n_1 C_v' dT$**. The increase in internal energy of the second gas** $= n_2 C_v'' dT$ **and the increase in internal energy of the mixture** $= (n_1 + n_2) C_v dT$**. Thus,**

$(n_1 + n_2) C_v dT = n_1 C_v' dT + n_2 C_v'' dT$

Or, $C_v = \frac{n_1 C_v' + n_2 C_v''}{n_1 + n_2}$ **(i)**

$$C_p = C_v + R = \frac{n_1 C_v' + n_2 C_v''}{n_1 + n_2} + R$$

$$= \frac{n_1 (C_v' + R) + n_2 (C_v'' + R)}{n_1 + n_2}$$

$$= \frac{n_1 C_p' + n_2 C_p''}{n_1 + n_2} \quad \text{(ii)}$$

From (i) and (ii), $\gamma = \frac{C_p}{C_v} = \frac{n_1 C_p' + n_2 C_p''}{n_1 C_v' + n_2 C_v''}$

$= \frac{4 \times \frac{5}{2}R + 2 \times \frac{7}{2}R}{4 \times \frac{3}{2}R + 2 \times \frac{5}{2}R} = \frac{17}{11}$

Q.15 Answer:

103 °C

Solution:

We are given that power of the machine,

$P = 10 \text{ kW} = 10^4 \text{ W}$

Time for which the machine is used, $t = 2.5 \text{ min} = 150 \text{ s}$

Mass of the aluminium block, $m = 8 \text{ kg} = 8 \times 10^3 \text{ g}$

Specific heat of aluminium, $c = 0.91 \text{ J/g}^\circ\text{C}$

Since 50% **of the energy is used up in heating the machine itself or lost to the surroundings,**

Energy absorbed by the block,

$Q = 0.5Pt = 0.5 \times 10^4 \times 150 \text{ J} = 7.5 \times 10^5 \text{ J}$

Let ΔT **be the rise in temperature of the block.**

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As $Q = mc\Delta T$

$\Rightarrow \quad \Delta T = \frac{Q}{mc} = \frac{7.5 \times 10^5 (\text{ J})}{8 \times 10^3 (\text{ g}) \times 0.91 (\text{ J/g}^\circ\text{C})} = 103^\circ\text{C}$

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